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From Emergence to Control: Probing and Modulating Self-Reflection in Language Models

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ProbingReflection

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Abstract

Self-reflection—the ability of a large language model (LLM) to revisit, evaluate, and revise its own reasoning—has recently emerged as a powerful behavior enabled by reinforcement learning with verifiable rewards (RLVR). While self-reflection correlates with improved reasoning accuracy, its origin and underlying mechanisms remain poorly understood. In this work, *we first show that self-reflection is not exclusive to RLVR fine-tuned models: it already emerges, albeit rarely, in pretrained models*. To probe this latent ability, we introduce Reflection-Inducing Probing, a method that injects reflection-triggering reasoning traces from fine-tuned models into pretrained models. This intervention raises self-reflection frequency of Qwen2.5 from 0.6% to 18.6%, revealing a hidden capacity for reflection. Moreover, our analysis of internal representations shows that both pretrained and fine-tuned models maintain hidden states that distinctly separate self-reflective from non-reflective contexts. Leveraging this observation, *we then construct a self-reflection vector, a direction in activation space associated with self-reflective reasoning*. By manipulating this vector, we enable bidirectional control over the self-reflective behavior for both pretrained and fine-tuned models. Experiments across multiple reasoning benchmarks show that enhancing these vectors improves reasoning performance by up to 12%, while suppressing them reduces computational cost, providing a flexible mechanism to navigate the trade-off between reasoning quality and efficiency without requiring additional training. Our findings further our understanding of self-reflection and support a growing body of work showing that understanding model internals can enable precise behavioral control.

1 Introduction

Reinforcement Learning with Verifiable Rewards (RLVR) has emerged as a powerful technique for enhancing the reasoning abilities of large language models (LLMs), enabling learning from outcome-level feedback across diverse tasks [1–3]. In a nutshell, RLVR optimizes for end-task success, allowing models to explore novel reasoning strategies at scale [4–6]. Notably, it has been reported [7–9] that such training induces new emergent behaviors, such as self-reflection—the ability of a model to revisit, evaluate, and revise its prior outputs. For instance, the DeepSeek-R1 report [7] highlights that RLVR-trained models often generate tokens such as "wait", interpreted as signals of internal deliberation or critique.

Despite these observations, it remains unclear why such reflective behaviors emerge. Answering this question is crucial for understanding the foundations of reasoning in LLMs and guiding future methods for enhancing their performance. Moreover, empirical studies have shown that reflection correlates with more accurate and robust reasoning [10–12], and that prompting models to explicitly "wait" or reflect can further improve performance with test-time computing. However, this benefit may come at a cost: reflection can increase inference time, introduce unnecessary verbosity, and reduce computational efficiency [13–15]. These findings highlight not only the need to understand self-reflection in LLMs, but also the importance of controlling it to balance reasoning quality and efficiency. These gaps motivate this work to study the following research questions:

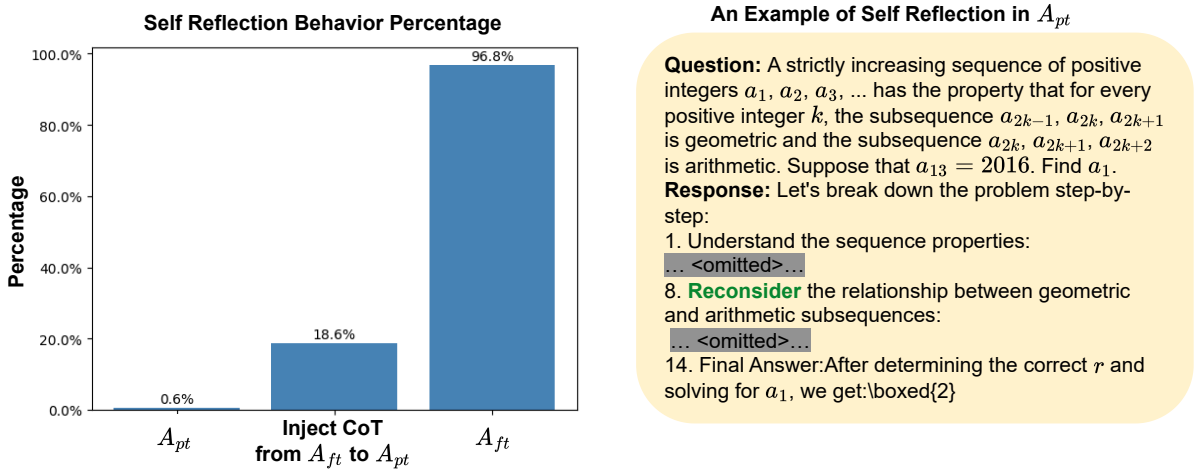

Figure 1: **Left:** Frequency distribution of self-reflection behaviors for pretrained model $\mathcal{A}_{\texttt{pt}}$, fine-tuned model $\mathcal{A}_{\texttt{ft}}$, and $\mathcal{A}_{\texttt{pt}}$ with reflection-inducing probing by injecting CoT from $\mathcal{A}_{\texttt{ft}}$, evaluated on the MATH500 dataset. **Right:** A representative example of spontaneous self-reflection in $\mathcal{A}_{\texttt{pt}}$, demonstrating that this capability emerges naturally during pretraining, albeit with different self-reflection tokens than those typically observed in RLVR fine-tuned models.

Is self-reflection a novel behavior induced by RLVR, or does it already emerge during pretraining? Can we control self-reflection in LLMs to balance performance and computational efficiency?

Contribution In this work, we provide affirmative answers to both questions. First, we compare the reasoning behaviors of pretrained models and fine-tuned models (either via RLVR or distillation), and verify that self-reflection is already present in the pretrained model, albeit at a much lower frequency. Next, we analyze the hidden representations associated with reflective versus non-reflective reasoning, and find that they exhibit distinct activation patterns. Furthermore, we show that the degree of self-reflection can be modulated by a single direction in the representation space. Our contribution can be summarized as follows.

- **Self-Reflection already emerge during pretraining:** We demonstrate that self-reflection capabilities naturally exist in pretrained models and are not solely artifacts of RLVR. However, the frequency of such behavior is extremely low—for example, only 0.6% as shown in Figure 1. To isolate the model's capacity for self-reflection from its general reasoning ability, we propose a method, called *reflection-inducing probing*, that inserts reasoning traces—specifically, those that trigger self-reflection in a fine-tuned reasoning model—into the input of the pretrained model, and then measures whether the latter produces reflection in response. Using reflection-inducing probing, we observe that the pretrained model exhibits reflection with a frequency of 18.6%, significantly higher than the baseline of 0.6%, though still lower than the fine-tuned model (which is almost 100%). Through comparative analysis of hidden representations, we show that pretrained models maintain internal structures that distinguish reflective behavior from non-reflective contexts—similar to fine-tuned models—further suggesting that pretrained models already possess self-reflection capabilities.
- **The degrees of self-reflection can be modulated by a single direction:** Motivated by the separability of reflective and non-reflective contexts in the hidden representation space, we use the method of difference-of-means [16] to construct a *self-reflection direction*, enabling control over self-reflection behavior for both pretrained and fine-tuned models. Our experiments demonstrate that this control mechanism offers a tunable trade-off between accuracy and efficiency: enhancing reflection improves accuracy by up to 12% on benchmarks, while suppressing it reduces output length by over 32% without significant performance degradation. We further show that this direction transfers robustly across diverse tasks—including mathematical and scientific reasoning—highlighting its universality as a shared, task-agnostic mechanism.

2 Preliminary

2.1 Transformer Layer

Decoder-only transformers [17] map input tokens $\mathbf{t} = (t_1, t_2, \ldots, t_n) \in \mathcal{V}^n$ to output probability distributions over the vocabulary $\mathcal{V}$. Let $\mathbf{h}_i^{(\ell)} \in \mathbb{R}^d$ denote the residual stream activation (also referred to as the hidden state) of the i-th token at the ℓ-th layer, where d is the dimensionality of the hidden state. Each of the L transformer layers applies a sequence of attention and MLP transformations to update the residual stream:

$$\tilde{\mathbf{h}}_i^{(\ell)} = \mathbf{h}_i^{(\ell)} + \text{Attn}^{(\ell)}(\mathbf{h}_{1:i}^{(\ell)}), \quad \mathbf{h}_i^{(\ell+1)} = \tilde{\mathbf{h}}_i^{(\ell)} + \text{MLP}^{(\ell)}(\tilde{\mathbf{h}}_i^{(\ell)}) \tag{1}$$

The final hidden state is then projected to a probability distribution over the vocabulary $\mathcal{V}$ using an unembedding matrix followed by a softmax function.

2.2 Self-Reflection

Recent work has shown that large language models (LLMs), even when pretrained purely on next-token prediction, demonstrate surprising levels of reasoning ability [18–20]. However, this capability can be significantly enhanced through fine-tuning on reasoning tasks using either reinforcement learning with verifiable rewards (RLVR) or supervised learning with distilled responses from reasoning models trained from RLVR [21–23]. We denote the pretrained model as $\mathcal{A}_{\texttt{pt}}$, and its fine-tuned variant as $\mathcal{A}_{\texttt{ft}}$.

A notable emergent behavior observed in fine-tuned models $\mathcal{A}_{\texttt{ft}}$ is self-reflection—the model's ability to internally evaluate, critique, or revise its own reasoning process. Unlike standard reasoning, which involves generating a direct solution to a task, self-reflection introduces an intermediate meta-cognitive step where the model pauses or backtracks to reconsider its prior outputs. This behavior is often marked by explicit tokens such as "wait," which have been shown to correlate with improved reasoning outcomes [8, 24, 25].

Importantly, self-reflection is not limited to any specific model architecture, observed across both proprietary models [26] and open-source systems [7, 27, 28], indicating that it may be a general emergent property of optimizing for complex reasoning objectives. While models may signal self-reflection using various phrases, including "wait", "let me double-check", or "I might have made a mistake", we focus our analysis on the canonical token "wait" due to its high frequency and clear association with reflective behavior. Our analysis confirms that "wait" is the most commonly used reflection marker across the DeepSeek-R1 series of models [7]. Supporting analyses can be found in Appendix B. Crucially, the analytical framework we develop can generalize beyond "wait", extending to any token that plays an analogous reflective role within the reasoning trajectory.

3 Self-Reflection Already Emerges During Pretraining

In this section, we conduct a systematic analysis of self-reflection behavior in both pretrained models $\mathcal{A}_{\texttt{ft}}$ and fine-tuned ones $\mathcal{A}_{\texttt{pt}}$. Using the MATH500 dataset [29] as our evaluation benchmark, we compare DeepSeek-R1-Distill-Qwen-1.5B ($\mathcal{A}_{\texttt{ft}}$), a model fine-tuned from Qwen2.5-1.5B ($\mathcal{A}_{\texttt{pt}}$) [7, 30]. First, we show that self-reflection naturally emerges in pretrained models, though at a substantially lower frequency than in RLVR-distilled counterparts. Second, through analysis of hidden state representations, we find that even pretrained models implicitly encode and differentiate between self-reflective and nonself-reflective states, despite rarely generating reflective outputs explicitly.

3.1 Probing Self-Reflection in Pretrained Models

To investigate whether self-reflection emerges intrinsically in $\mathcal{A}_{\texttt{pt}}$, rather than being solely a byproduct of fine-tuning strategies such as RLVR, we examined the behavior of $\mathcal{A}_{\texttt{pt}}$ on mathematical reasoning tasks using the MATH500 benchmark. Figure 1 (left) highlights the contrast in self-reflection frequency between $\mathcal{A}_{\texttt{pt}}$ and $\mathcal{A}_{\texttt{ft}}$ models, while the right panel shows a representative instance of naturally occurring self-reflection in $\mathcal{A}_{\texttt{pt}}$. Remarkably, even in the absence of task-specific supervision, $\mathcal{A}_{\texttt{pt}}$ exhibited spontaneous self-reflective behavior in a small but non-negligible fraction of cases—approximately 0.6%, as shown in Figure 1. These